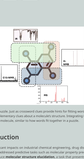
MolPuzzle

Can LLMs Solve Molecule Puzzles? A Multimodal Benchmark for Molecular Structure Elucidation

Kehan Guo[1*], Xiuzhen Yan[1*], Yanbang Cui[2], Zhixun Li[2], Zhixun Guo[2], Mihir Shome[1], Zhenwen Liang[1], Nitesh V. Chawla[1], Olaf Wiest[1], Xiangliang Zhang[1] ✉

[1]Department of Computer Science and Engineering, University of Notre Dame.
[2]Department of Chemistry and Biochemistry, University of Notre Dame.
✉ email: kguo@nd.edu, olaf.wiest@nd.edu, xzhang1@nd.edu

▶ Slides ▶ Video ▶ Blog ▶ Twitter



Comparison of molecular structure elucidation to solving a crossword puzzle. Just as crossword clues provide hints for filling words into a grid, spectroscopic data provides clues for assembling molecular structures. MolPuzzle simulates this process through multi-modal spectra interpretation, iterative structure assembly, and feedback-driven refinement, offering a framework for AI-assisted molecular structure elucidation.

Introduction

Artificial intelligence (AI) is revolutionizing chemistry, with significant impacts on industrial chemical engineering, drug discovery, and education. Large language models (LLMs) have successfully addressed molecular tasks such as molecular property prediction, molecule captioning, and text-based molecule generation. However, the critical process of elucidating molecular structures from spectroscopic data remains a challenge for AI. **This task requires integrating diverse spectroscopic data, iterative hypothesis testing, and deep chemical reasoning to determine a molecule's structure.** Much like solving a complex crossword puzzle, it involves piecing together clues from various spectra to assemble the final molecular "puzzle". Addressing the limitations in strategy and complexity between molecular structure elucidation and solving a crossword puzzle,

In this work, we present a novel approach to molecular structure elucidation, adapting the task for Large Language Models (LLMs). **We introduce MolPuzzle, a comprehensive framework that includes SMI-MMS, a multi-modal dataset for LLM training and evaluation comprising 234 complete structure elucidation challenges involving multimodal data (the 1D NMR, 2-HMR, and 1-MHR spectra, as well as molecular formulas.** Each instance requires LLMs to navigate three key sub-tasks: molecular understanding, spectrum interpretation, and molecule construction.

We extend 11 state-of-the-art LLMs, including GPT-4o and Claude 3 Opus, alongside human benchmarks, like Mistral-Instruct 22B and Llama-3-Instruct 70B, on these challenges using zero-shot prompting and our proposed 5-step workflow. **Our results show that while top-performing LLMs demonstrate promising capabilities, they still lag significantly behind human expert performance, primarily stemming the ground truth(G-2-G) score critical for spectrum interpretation and molecule construction.**

In summary, our contributions are twofold. Our contributions are twofold: **(1) A rice reasoning challenge for the AI community focused on complex problem-solving in chemistry and (2) Rate of hulls for the chemistry community, showcasing LLM potential to accelerate molecular structure elucidation and inspire interdisciplinary collaboration.**

Overview of the MolPuzzle Benchmark

The MolPuzzle benchmark is designed to test the reasoning capabilities of Large Language Models (LLM) in molecular structure elucidation. In this work, we propose a 5-step workflow that addresses three core challenges: molecular understanding, simulating real-world chemistry tasks, and evaluating the quality of multi-step reasoning.

- **Molecule Understanding.** This stage evaluates the model's ability to identify and understand basic molecular structures, starting from the molecular formula derived from mass spectrometry data. The dataset includes questions about the degree of unsaturation, aromatic rings, and functional groups, helping the model narrow down possible molecular structures.
- **Spectrum Interpretation.** This stage assesses the model's capability to interpret 1D and 2D NMR spectra, which provide crucial clues about molecular fragments and their connectivity. These spectral images provide critical information about functional groups, molecular mass, and the arrangement of atoms. The dataset challenges models to integrate these clues and refine molecular hypotheses based on the spectral data.
- **Molecule Construction.** The final stage tests the model's ability to assemble the molecular structure using the fragments hypothesized from previous steps. This involves combining a pool of molecular fragments to form a complete molecular structure that aligns with all the spectroscopic data and satisfies the initial molecular formula.

1. Identify molecule substructures based on molecule formula

Prompt: As an expert organic chemist, your task is to analyze the provided molecular formula and identify potential molecular structures and the functional groups within them. Leverage your knowledge to systematically explore and identify plausible molecular substructures.

Answer: Carbonyl(O bond) (Ring) degree of unsaturation = 1

2. Refine the substructure pools based on Spectrum images.

3. Select fragments from the pools to assemble the molecule SMILES and validate

Prompt: Selected one fragment from the list of fragments to add to the molecule. Ensure the chosen fragment is a valid chemical group that can be attached to the existing molecule. Update the molecular structure and calculate the unsaturation score (U.S) resulting from adding this fragment.

Answer: Based on the suggestions, the fragment C(=O)O is added to the current molecule. The updated molecular structure is represented by the SMILES C1=CC=C(C=C1)C(=O)O. The unsaturation score (U.S) remains 5, indicating the addition of the carboxyl group does not alter the unsaturation score in this case.

Stop: when run out of heavy atoms.

Answer: C1C(C(O)(C=O)O)O(C=O)C=C1O

(a) Stage 1 **(b) Stage 2** **(c) Stage 3**

In total, Molpuzzle includes 234,076 data examples collected from each Stage.

Statistic	Number
Total MolPuzzle Instances	234
Stage 1 QA samples	7,020
- Molecular formula	234
- Max question length	128
- Average question length	124.7
Stage 2 QA samples	11,502
- Max question length	128
- Max answer length	128
- Average question length	264
- Average answer length	8.13
- Maximum Rounds	5
- Max question length	256
- Average question length	216

Figure 3. Statistics of the MolPuzzle dataset

Figure 4. Inner ring: sample distribution in 3 stages. Outer ring: sample distribution across categories in each stage. SE: saturation identification, SAC: saturation atom count, FG: functional group identification, IAS: isomer atom-ring identification, S.A: spectrum analysis.

Experiment Results

Addressing individual QA tasks in three stages

We first conducted evaluation of a variety of LLMs for completing the individual tasks in each stage, including GPT-4o, GPT-3.5-turbo, Claude-3-opus, Gemini-pro, Llama-3-8B-Instruct, Mixtral-8x7B-Instruct-v0.1, and in particular multimodal LLMs such as Gemini-pro-vision, Llama-Vision-70B, Qwen-VL-Chat, and Intern2-VL-26B-Chat.

Method	Stage 1 Molecule Understanding Tasks			
	SI	AN	FG	IDC
GPT-4o	**1.00±0.000**	0.562±0.014	0.334±0.005	0.667±0.016
GPT-3.5-turbo	0.87±0.025	0.316±0.011	0.203±0.005	0.5±0.000
Claude 3 Opus	0.787±0.005	**0.566±0.014**	**0.514±0.001**	**0.850±0.014**
Gemini	0.276±0.011	0.086±0.005	0.121±0.003	0.309±0.008
Mixtral	**1.00±0.000**	0.366±0.005	0.295±0.009	0.871±0.000

Method	Stage 2 Spectrum Interpretation Tasks			
	2D Interpretation	MASS Interpretation	1-NMR Interpretation	2-NMR Interpretation
GPT-4o	**0.404±0.012**	**0.899±0.005**	**0.318±0.016**	**0.208±0.016**
Gemini	0.154±0.025	0.151±0.021	0.116±0.008	0.204±0.015
Human	0.556±0.031	0.786±0.047	0.75±0.05	0.583±0.031

Method	Stage 3 Molecule Construction Tasks	
	1-NMR Elucidation	2-NMR Elucidation
GPT-4o	**0.334±0.010**	**0.504±0.007**
Gemini	0.245±0.015	0.352±0.017
Human	0.75±0.05	0.792±0.042

Table 1. F1 scores (±1 of Individual QA tasks in three stages. The best LLM's results are in bold font. Axis.

Table 2: Stage 1 (SI: Saturation Identification, AN: Atomic Ring Identification, FG: Functional Group Identification, and IDC: Saturation Degree Calculation).

Addressing entire molecule puzzles

For solving the entire molecule puzzle, the evaluation is limited to the three most advanced multimodal LMMs: GPT-4o, Claude-3-opus, and Gemini-pro, due to the incorporation of spectrum image analysis in Stage 2.

Method	Jac. (↑)	Levenshtein (↓)	MAE(↓)	MAPE (↓)	Morgan FTS (↑)
GPT-4o	**0.014±0.004**	**17.455±0.375**	**1.993±0.082**	**0.411±0.030**	0.293±0.012
Claude-3-opus	0.013±0.008	19.606±0.386	**1.993±0.082**	0.493±0.034	0.249±0.007
Gemini-pro	0.002±0.000	19.715±0.196	**1.993±0.082**	0.509±0.017	0.094±0.002
Human	0.667±0.047	5.92±0.375	**1.993±0.082**	**0.409±0.023**	0.878±0.150

Table 3: The performance of LLMs and human baseline in solving MolPuzzle. The best LLM results are in bold font. Acc. stands for the Accuracy of Exact Match.

Success and Failure Analysis

Error in solving the molecule puzzle

The Figure presents case studies that illustrate the challenges faced by LLMs in synthesizing the more complex errors made by GPT-4o. **the accumulation of errors in iterative steps, which can lead to a compounding effect where initial errors in the process** can identify the correct fragments and assembling them step by step to form the final molecular structure. We find that **GPT-4o effectively leverages the provided molecular fragments and demonstrates a strong understanding of chemical rules. However, as the process progresses, it does so select structures that satisfy all the requirements indicated by the NMR data.**

BibTeX

```
@article{anonymous2024molpuzzle:
MolPuzzle: Can Large Language Models Solve Molecule Puzzles? A Multimodal Benchmark for Molecular Structure Elucidation,
Author=Guo, Kehan and Yan, Xiuzhen and Cui, Yanbang and Yan, Zhixun and Guo, Zhixun and Shome, Mihir and Liang, Zhenwen and Chawla, Nitesh V. and Wiest, Olaf and Zhang, Xiangliang,
Submitted=The Thirty-eight Conference on Neural Information Processing Systems Datasets and Benchmarks Track(2)
```

UNIVERSITY OF NOTRE DAME
NSF
Grant Nos. 2340651 & 2340652
C-CAS
NSF Center for Computer Assisted Synthesis

This website is website adapted from Nerfies, licensed under a Creative Commons Attribution-ShareAlike 4.0 International License.